🧪 Digital Forensics Case Study

Insider Intellectual Property Theft – ShockWave Analytics

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📘 Case Background

ShockWave Analytics is a mid-sized cybersecurity software company specialising in predictive threat detection tools for financial institutions. The company has been developing a new proprietary machine-learning engine:*Project Triton*, expected to generate significant competitive advantage.

John Matthews, a senior software engineer, had been employed at ShockWave for five years. He was one of four developers with full access to the Triton source code repository and design documentation.

Timeline of Events

- **March 3** – John submits formal resignation, stating he accepted a position at a "different industry startup."
- **March 17** – John's final working day.
- **March 28** – A competitor, BasilCore Systems, announces a new product with architecture remarkably similar to Triton.
- **April 2** – ShockWave's internal review discovers unusual file access patterns in the two weeks before John's departure.

Senior management suspects potential intellectual property theft and contacts the legal department.
A formal digital forensic investigation is initiated to determine:

- Whether proprietary information was stolen
- Whether John acted alone or in collaboration
- Whether the company has grounds for civil litigation

ShockWave IT preserved:

- *John's company-issued laptop, A USB storage device found in his desk drawer, Relevant network logs, Email server archives, File server access logs, HR resignation and termination documentation*

You are assigned as forensic analysts engaged by ShockWave's legal team.
The task is to conduct a forensic analysis

Evidence Sources Available

Forensic image of John's company laptop

Forensic image of USB storage device recovered from his desk

Corporate firewall logs

Email server logs and mailbox archive

Windows Security and System Event Logs from the laptop

File server access logs (Project Triton directory)

HR resignation and exit timeline documentation